

3 Demand Theory—Applications

We are now prepared to apply the demand theory developed in chapter 2 to different settings. First, we examine the different measures that one can use to quantify the welfare change that results from a price change. Intuitively, if a good becomes cheaper consumers are better off, but by how much? Answering this question depends on the tool from demand theory that we use, be it Walrasian or Hicksian demand. In using Walrasian demand $x(p, w)$, we could measure the welfare increase by the area below the demand curve and between the initial and final price, which is the consumer surplus. If instead we use Hicksian demand $h(p, u)$, we must first specify which utility level we use in the argument, that is, the utility level that the individual reaches before or after the price decrease. Using one or the other has consequences in the welfare measure that we will obtain. In particular, if we use the utility level before the price change, we obtain a welfare measure commonly referred to as the “compensating variation,” whereas if we use the utility level after the price change, we find the “equivalent variation.” Importantly, these welfare measures do not necessarily coincide, but as sections 3.3 and 3.4 show, the measurement error is small (allowing us to use either measure without major problems) when the price change is small and/or when the income effect associated to such a price change is small. A special type of preference relation exhibiting zero income effects is quasi-linear preferences, which thus gives rise to the same welfare measure when using consumer surplus, compensating and equivalent variation.

Sections 3.5 through 3.7 take a more applied focus. First, we consider a more compact representation of the Slutsky equation that allows for direct tests in terms of elasticities. Second, we apply the income and substitution effect to labor economics (where the change in price is now a change in hourly wages). This analysis helps us apply the Slutsky equation to labor markets, and subsequently examine under which conditions a decrease in income tax rates can yield an increase in total tax collection, as prescribed by the Laffer curve. Third, we extend our analysis of the income and

substitution effect (where we examined how a cheaper good affects its own demand) to settings with multiple goods. In particular, we explore whether a lower price for a good x increases the Walrasian demand for a different good y (implying that goods x and y are regarded as gross substitutes), or whether it increases the Hicksian demand of good y (and thus goods are net substitutes). We then elaborate on the connection between the gross and net substitutability property, as the former does not necessarily imply the latter. Finally, section 3.8 discusses demand aggregation (i.e., market demand) where we investigate under which conditions it can be represented as a function of aggregate wealth. As we describe in that section, this is possible if the indirect utility function of all individuals can be represented with the so-called Gorman form function.

3.1 Measuring the Welfare Effects of a Price Change

In this section we aim at measuring the welfare effects that arise from a price increase or decrease or, alternatively, from the effects of introducing a tax or a subsidy that modifies the prices of some goods. Figure 3.1 represents a decrease in the price of x_1 , from p_1 to p'_1 yielding an increase in the utility level that the consumer reaches after solving his UMP, from u^0 to u^1 .

A natural question at this point is: Why not use the difference in the individual's utility level, $u^1 - u^0$, as a measure of welfare change? Such difference, however, exhibits two problems. First, from a within-subject approach, the difference $u^1 - u^0$ is not informative. Indeed, since only the ranking between two different bundles (ordinality) matters, we are only interested in whether $u^1 > u^0$ or $u^1 < u^0$, but not the difference.¹ Second, from a between-subjects approach, the utility measure $u^1 - u^0$ would not be comparable among different individuals with distinct utility functions. In order to avoid both of these criticisms, we develop monetary evaluations that measure the welfare change associated to a price change. In particular, for a decrease in the price of good 1, from p_1^0 to p_1^1 , we can evaluate the change in consumer welfare by using either of the following measures:

1. *Compensating variation (CV)* This welfare measure responds to the question: How much money would a consumer be willing to give up *after* a reduction in

¹ Recall that the utility function is subject to strictly increasing transformations that represent the same underlying preference relation. Hence the application of a monotonic transformation could increase the distance $u^1 - u^0$ for a given decrease in prices, leading us to believe that the welfare gain associated to such a price decrease is larger.

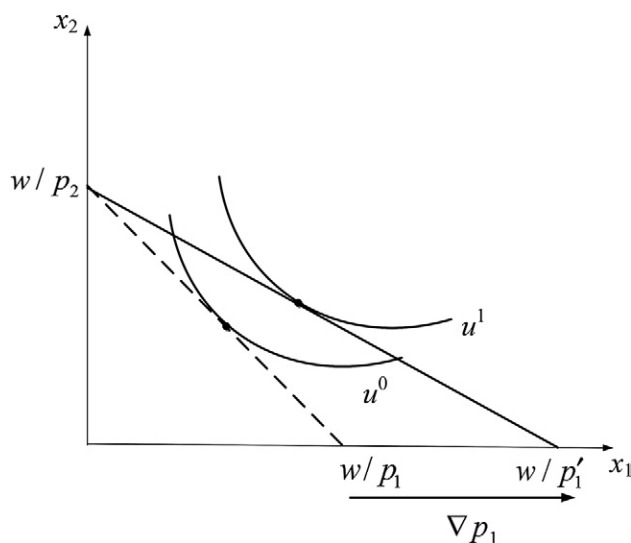


Figure 3.1
Measuring the welfare effects of a price decrease

prices to be just as well off as *before* the price decrease? For this reason we refer to this measure as the “compensating variation,” since it represents the change in income that would exactly compensate the consumer for the impact that the price change has on his welfare. Importantly, this measure maintains the initial utility level (before any price change), u^0 .

2. *Equivalent variation (EV)* It responds to the question: How much money would a consumer need *before* a reduction in prices to be as well off as *after* the price decrease? For this reason we refer to this measure as the “equivalent variation,” since it reflects the change in income that would be equivalent to the impact that the price change has on his welfare. In this case we focus on the final utility level that the individual can reach after the price change, u^1 .

3.1.1 CV and EV Using the Expenditure Function

Let us first examine the *CV*. Note that we can formalize the above intuition using either the expenditure function or the Hicksian demand. Using the expenditure function, we have

$$CV(p^0, p^1, w) = e(p^1, u^1) - e(p^1, u^0).$$

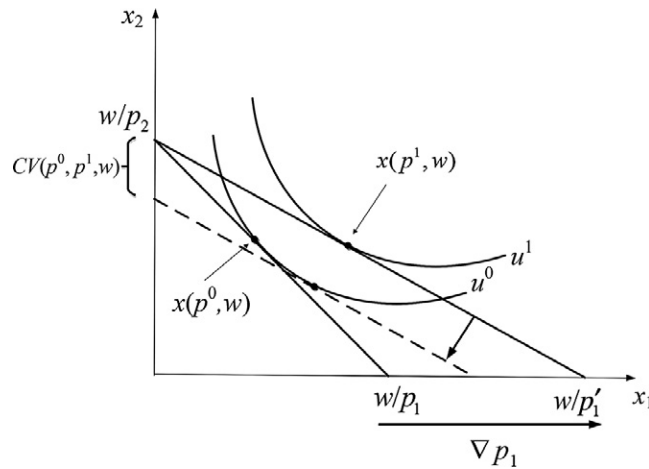


Figure 3.2
Compensating variation

Intuitively, this expression represents the amount of money that the consumer is willing to give up *after* the price decrease (when the price level is p^1 and his utility level has improved to u^1), to be just as well off as *before* the price decrease (when reaching the lower utility level u^0). Figure 3.2 represents the compensating variation using the expenditure function.

Here is a basic procedure on how to find the CV from a price decrease. Before the price change, the consumer selects optimal bundle $x(p^0, w)$, reaching utility level u^0 under budget set $B_{p^0, w}$. After p^1 decreases, the consumer chooses an optimal bundle $x(p^1, w)$ under budget set $B_{p^1, w}$. We then adjust the consumer's wealth (after the price change) to make him just as well off as before the price change. We represent this adjustment by a parallel shift in budget line $B_{p^1, w}$ to the dashed budget line (where the parallel displacement implies that we maintain the final price ratio, p^1 , but reduce his wealth), where the individual can reach his initial utility level u^0 at the final price ratio.² At this point his expenditure is $e(p^1, u^0)$. Hence the difference in expenditure $e(p^1, u^1)$ —at $B_{p^1, w}$ —and $e(p^1, u^0)$ —at the dashed budget line—represents the compensating variation resulting from the price change.

Remark If the price of good 2, p_2 , is normalized to 1 (or, more generally, if x_2 represents a composite commodity embodying all other goods different from x_1), we can

² You probably recognized that the wealth compensation we applied in the CV coincides with the so-called Hicksian wealth compensation defined in chapter 2, since we adjust the wealth level of the individual after the price change for him to reach his initial utility level.

then depict the size of the compensating variation in the vertical axis of figure 3.2. Graphically, since the price of the good represented in the vertical axis is \$1, the height of the vertical intercept of the budget line represents the wealth level needed to buy any bundle on that budget line. The compensating variation is, hence, the difference between the final budget after the price change, measured by the total expenditure on budget line $B_{p^1, w}$, and the consumer's wealth after the compensation, measured by the vertical axis of dashed budget line in figure 3.2.

Let us now turn to the equivalent variation (EV). Using the expenditure function, we can define EV as follows:

$$EV(p^0, p^1, w) = e(p^0, u^1) - e(p^0, u^0).$$

Intuitively, the EV represents the amount of money that the consumer is willing to receive *instead of* the price decrease. That is, at the initial price level p^0 (when his utility level is still u^0) we find the monetary amount that makes him as well off as after the price decrease (when he reaches the higher utility level u^1). Figure 3.3 illustrates the EV in a figure similar to figure 3.2.

Like for the CV , here is a basic procedure on how to find the EV from a price decrease. Before the price change, the consumer selects optimal bundle $x(p^0, w)$, reaching utility level u^0 . After p_1 decreases, the consumer chooses an optimal bundle $x(p^1, w)$ under budget set $B_{p^1, w}$. (Note that until this point we are just replicating the same analysis as for the CV .) Using the definition of the EV , we now must increase the initial

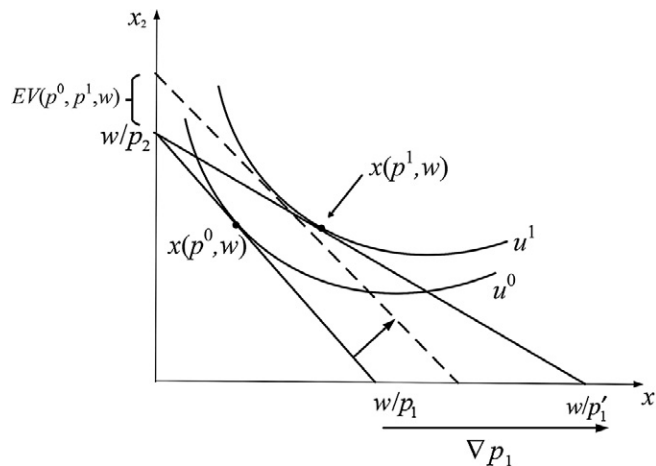


Figure 3.3
Equivalent variation

wealth of the individual (before the price change), $B_{p^0, w}$, to make sure he is as well off as after the price change. Graphically, we do so by upward shifting the initial budget line in a parallel fashion. Intuitively, we are increasing the initial wealth of the individual so that, at the initial price ratio, he can still reach utility level u^1 (the high utility level he can obtain after the price decrease). Therefore the difference in expenditure $e(p^0, u^1)$ —at the dashed budget line—and $e(p^0, u^0)$ —at $B_{p^0, w}$ —represents the equivalent variation associated with the price change. That is, EV represents the amount of money needed to obtain the new utility at the old prices.³

3.1.2 CV and EV Using Hicksian Demand

In our previous analysis we used the expenditure function to measure both the CV and EV . Nonetheless, both expressions can also be defined using the Hicksian demand. First, the expression of the CV can also be rearranged as follows (recall that only p_1 decreases, from p_1^1 to p_1^0 , while the price of all other goods remains constant, in that $p_k^0 = p_k^1$ for all other goods $k \neq 1$):

$$\begin{aligned} CV(p^0, p^1, w) &= \underbrace{e(p^1, u^1)}_w - e(p^1, u^0) \\ &= w - e(p^1, u^0) \\ &= \underbrace{e(p^0, u^0)}_w - e(p^1, u^0) \\ &= \int_{p_1^1}^{p_1^0} \frac{\partial e(p_1, \bar{p}_{-1}, u^0)}{\partial p_1} dp_1 = \int_{p_1^1}^{p_1^0} h_1(p_1, \bar{p}_{-1}, u^0) dp_1, \end{aligned}$$

since $e(p^1, u^1) = e(p^0, u^0) = w$ and $\partial e(p, u) / \partial p_1 = h_1(p_1, u)$. In addition let $\bar{p}_{-1}$ denote the price vector of all other goods $k \neq 1$ such that $\bar{p}_{-1} = (p_2, p_3, \dots, p_L)$, which are unaffected by the price change. Since $p_1^1 > p_1^0$, the CV of a price decrease is negative, thus indicating the amount of money that the individual would be willing to give up after the price decrease in order to be as well off as before the price decrease.

Therefore the CV can be represented by the area to the left of the Hicksian demand curve for good 1, associated with utility level u^0 , and lying between the initial price p_1^0 and the final price p_1^1 , as figure 3.4 indicates. In the figure note that the price decrease helps the consumer move from utility level u^0 to u^1 , and therefore from Hicksian demand $h_1(p_1, \bar{p}_{-1}, u^0)$ to $h_1(p_1, \bar{p}_{-1}, u^1)$. However, the CV implies that we need to adjust

³ Finally, note that if $p_2 = \$1$ we can depict the equivalent variation in the vertical axis of figure 3.3. In particular, the equivalent variation is measured by the difference between the amount of wealth left to the consumer after the wealth compensation, measured by the dashed budget line, and the initial budget before the price change, measured by $B_{p^0, w}$, as indicated in the vertical axis of the figure.

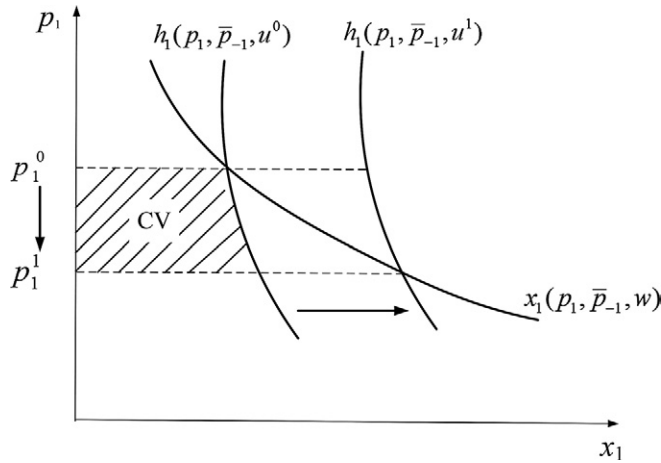


Figure 3.4
Compensating variation using $h(p, u)$

the final wealth level of the individual such that, after the price decrease, he reaches his initial utility level u^0 , thus remaining at $h_1(p_1, \bar{p}_{-1}, u^0)$. Hence the welfare gain resulting from the price decrease is represented by the shaded area in the figure.⁴

Operating in a similar fashion, let us now represent the expression of the EV using the Hicksian demand curve. If p_1 decreases, meaning $p_1^1 < p_1^0$ and $p_k^1 \neq p_k^0$, for all other goods $k \neq 1$, the EV is

$$\begin{aligned}
 EV(p^0, p^1, w) &= e(p^0, u^1) - \underbrace{e(p^0, u^0)}_w \\
 &= e(p^0, u^1) - w \\
 &= e(p^0, u^1) - \underbrace{e(p^1, u^1)}_w \\
 &= \int_{p_1^1}^{p_1^0} \frac{\partial e(p_1, \bar{p}_{-1}, u^1)}{\partial p_1} dp_1 = \int_{p_1^1}^{p_1^0} h_1(p_1, \bar{p}_{-1}, u^1) dp_1,
 \end{aligned}$$

where $e(p^0, u^0) = e(p^1, u^1) = w$ and $\partial e(p, u) / \partial p_1 = h_1(p_1, u)$. Hence the EV is represented by the area to the left of the Hicksian demand curve, associated with utility level u^1 , and lying between the initial price p_1^0 and the final price p_1^1 , as illustrated in figure 3.5.

⁴ Recall from chapter 2 that the Walrasian demand is flatter than the Hicksian demand curve when goods are regarded as “normal.” The converse holds if goods are regarded as “inferior.” Also remember that we represent the welfare change associated with a price decrease. The graphical representation of the CV resulting from a price increase differs from that depicted in figure 3.4 (we discuss this point below).

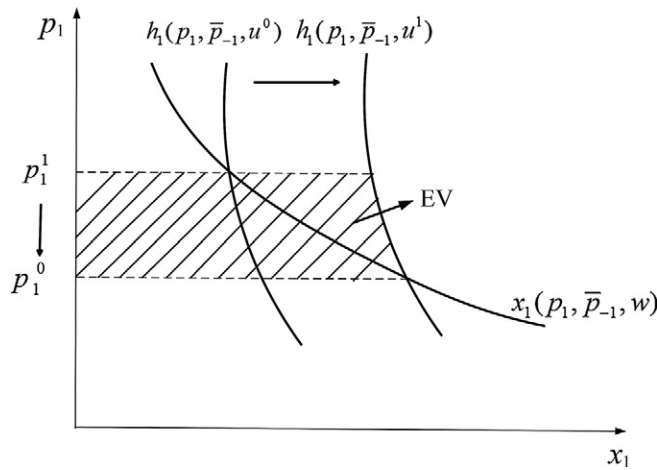


Figure 3.5
Equivalent variation using $h(p, u)$

Recall from our previous discussion that the price decrease helps the consumer move from utility level u^0 to u^1 , and therefore from Hicksian demand $h_1(p_1, \bar{p}_{-1}, u^0)$ to $h_1(p_1, \bar{p}_{-1}, u^1)$ in figure 3.5. The definition of the EV emphasizes that we need to adjust the initial wealth of the individual such that, under the initial price ratio, he can reach his final utility level u^1 and remain at $h_1(p_1, \bar{p}_{-1}, u^1)$. Graphically, his welfare gain is measured by the shaded area in the figure.⁵

We just discussed the analysis of price decreases. But what if we analyze a price increase? First, the Hicksian demand associated with the initial utility level u^0 (before the price increase, or before the introduction of a sales tax that increases the price of that good) experiences an inward shift, since now the consumer's utility level is u^1 , where $u^1 < u^0$. Hence $h_1(p_1, \bar{p}_{-1}, u^0) > h_1(p_1, \bar{p}_{-1}, u^1)$, so graphically the Hicksian demand now shifts inward as figures 3.6 depicts, as opposed to outward in the case of a price decrease, as depicted in figures 3.4 and 3.5. Therefore the definitions of the CV and EV change⁶:

1. CV The amount of money that a consumer would need *after* a price increase to be as well off as *before* the price increase.

⁵ Note that figure 3.5 assumes that x_1 is regarded as “normal” rather than “inferior” since the Walrasian demand is flatter than the Hicksian demand. See Chapter 2 for more details.

⁶ In order to facilitate the comparison with our definition of CV and EV in the event of a price decrease, we italicized the only elements that have changed.

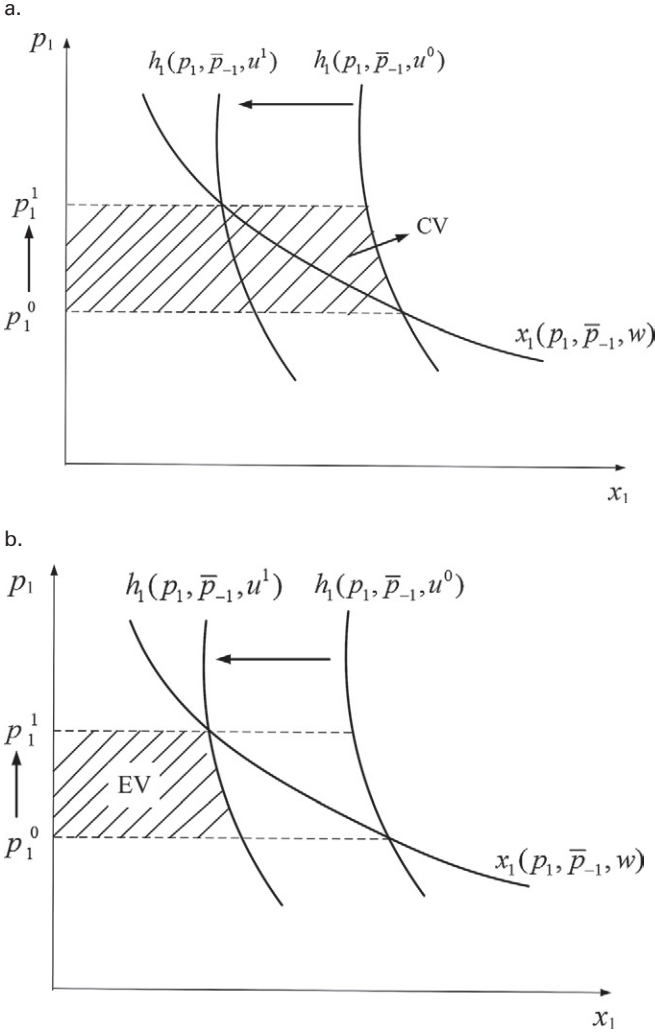


Figure 3.6
CV and EV of a price increase

2. *EV* The amount of money that the consumer would be willing to give up *before* the price increase to be as well off as *after* the price increase.

Let us now graphically represent the *CV* and *EV* areas associated to a price increase; see figure 3.6. At first glance, it seems that the labels of the *CV* and *EV* areas have been reversed! However, such relabeling is understandable: recall that the *CV* is associated to the area below the Hicksian demand $h_1(p_1, \bar{p}_{-1}, u^0)$, given that the *CV* always takes the initial utility level u^0 as a reference. Graphically, $h_1(p_1, \bar{p}_{-1}, u^0)$ now lies to the right of $h_1(p_1, \bar{p}_{-1}, u^1)$, since the price increase entails $u^1 < u^0$. In contrast, the *EV* is associated to the area below the Hicksian demand $h_1(p_1, \bar{p}_{-1}, u^1)$, meaning *EV* takes the final utility level u^1 as a reference. In this case $h_1(p_1, \bar{p}_{-1}, u^1)$ lies to the left of $h_1(p_1, \bar{p}_{-1}, u^0)$, since the price increase produced $u^1 < u^0$.⁷ (Note that a similar relabeling applies if we measure the *CV* and *EV* of a price increase using the expenditure function rather than the Hicksian demand.)

3.2 Measuring the Welfare Change Associated with the Introduction of a Tax

Our previous discussion of the welfare changes associated to a price increase can be applied to the analysis of the welfare effects of introducing a sales tax that increases the price of a good, such as alcohol or tobacco. This analysis, however, presents a difference from the previous section: we are interested in the area of the *CV* and *EV* that is *not* connected to tax revenue, since such revenue represents a sheer transfer of resources to another agent (the government). Hence we need to subtract the tax revenue from the expressions of the *CV* and *EV* used above to analyze price increases.

Specifically, when imposing a sales tax that increases p_1 , from p_1^0 to $(p_1^0 + t)$, the government collects a total tax revenue of

$$T = [(p_1^0 + t) - p_1^0] \cdot h_1(p_1, \bar{p}_{-1}, u^0) = t \cdot h_1(p_1, \bar{p}_{-1}, u^0) \quad \text{if using } CV,$$

$$T = [(p_1^0 + t) - p_1^0] \cdot h_1(p_1, \bar{p}_{-1}, u^1) = t \cdot h_1(p_1, \bar{p}_{-1}, u^1) \quad \text{if using } EV.$$

Figure 3.7 depicts the welfare loss associated with the introduction of this sales tax, where the welfare loss is measured using the *CV*. Similarly as we described above, the *CV* is represented by the area to the left of $h_1(p_1, \bar{p}_{-1}, u^0)$. However, not all of the welfare that consumers lose can be computed as a loss for the entire economy, since the government collects tax revenue T , as represented by the shaded rectangular area.

⁷ The *CV* is larger than the *EV* when p_1 increases, while the *CV* is smaller than the *EV* when p_1 decreases. We summarize our findings about the *CV* and the *EV* below. For applications of the equivalent and compensating variation to specific functional forms, see exercises 2 through 8 at the end of this chapter.

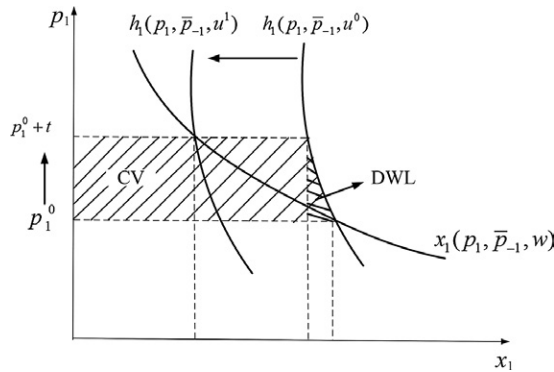


Figure 3.7
Welfare effect of a tax (*CV*)

There is nonetheless a portion of the *CV* that cannot be interpreted as a transfer from the consumer to the government in the form of tax revenue. Indeed the dark shaded triangular area immediately adjacent to $h_1(p_1, \bar{p}_{-1}, u^0)$ illustrates the net loss for the economy. We denote such a loss as the deadweight loss (*DWL*) of taxation and define it as follows⁸:

$$DWL = CV - T.$$

Similarly, if we use the *EV* to measure the welfare loss associated to the introduction of a tax, we have that the *DWL* of taxation is represented by the triangular area to the left of $h_1(p_1, \bar{p}_{-1}, u^1)$, as figure 3.8 indicates. In particular, the *EV* is captured by $EV(p^0, p^1, w) = \int_{p_1^0+t}^{p_1^0} h_1(p_1, \bar{p}_{-1}, u^1) dp_1$, and tax revenue is represented by $T = [(p_1^0 + t) - p_1^0] \cdot h_1(p_1, \bar{p}_{-1}, u^1)$, then the *DWL* from taxation is given by the difference

$$DWL = EV - T.$$

3.3 What If We Use the Walrasian Demand to Measure Welfare Changes?

The Walrasian demand is easier to observe than the Hicksian demand, and therefore it is tempting to use it in our measurements of welfare change. In particular, if we

⁸ In your public finance classes you probably used the triangular area to the left of the Walrasian, not Hicksian, demand curve in order to measure the *DWL* of taxation. Importantly, doing so when dealing with normal goods (inferior goods) yields a larger (smaller, respectively) *DWL*, relative to the measure provided by the *CV*. Below we elaborate on the consequences of using the Walrasian demand when measuring welfare changes.

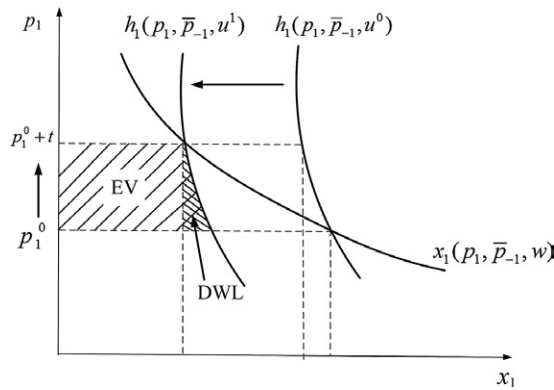


Figure 3.8
Welfare effect of a tax (*EV*)

consider the Walrasian demand, a decrease in p_1 would imply an increase in consumer surplus (*CS*).⁹ This measurement coincides with that provided by the *CV* and *EV* when income effects are zero. Indeed recall that the Walrasian demand measures both income and substitution effects resulting from a price change, whereas the Hicksian demand only captures substitution effects. Hence, there will be a difference between the *CV* and *CS*, and between the *EV* and *CS*. Figure 3.9 illustrates the difference between these welfare measures for the case in which good 1 is a “normal” good, namely income effects are positive. Hence the Walrasian demand is more sensitive to price changes than the Hicksian demand, implying that $CV < CS$ whereas $EV > CS$, meaning $EV > CS > CV$.

Example 3.1: Numerical example on *CV*, *EV*, and *CS* For a numerical example in which welfare change measured by the *CV*, *EV*, and *CS* does not coincide, consider a Cobb–Douglas utility function $u(x, y) = (x^{1/2})(y^{1/2})$, a wealth level of $w = \$100$, the price of good y normalized to \$1. If the price of good x is p_x , its Walrasian demand is given by $x(p_x, 100) = 50/p_x$. Assume now that the price of good x decreases from \$4 to \$3. In this context, the increase in consumer surplus is given by $CS = 34.65$. In order to measure the *CV*, we first find the initial bundle $x(p_0, w)$, the final bundle $x(p_1, w)$, and bundle $x(p_1, w')$ where w' denotes the adjusted income so the individual can reach his initial utility level at the final price ratio. It is straightforward to show that $x(p_0, w) =$

⁹ Other textbooks refer to the “change in consumer surplus” as the area variation (*ΔV*) associated to a price change.

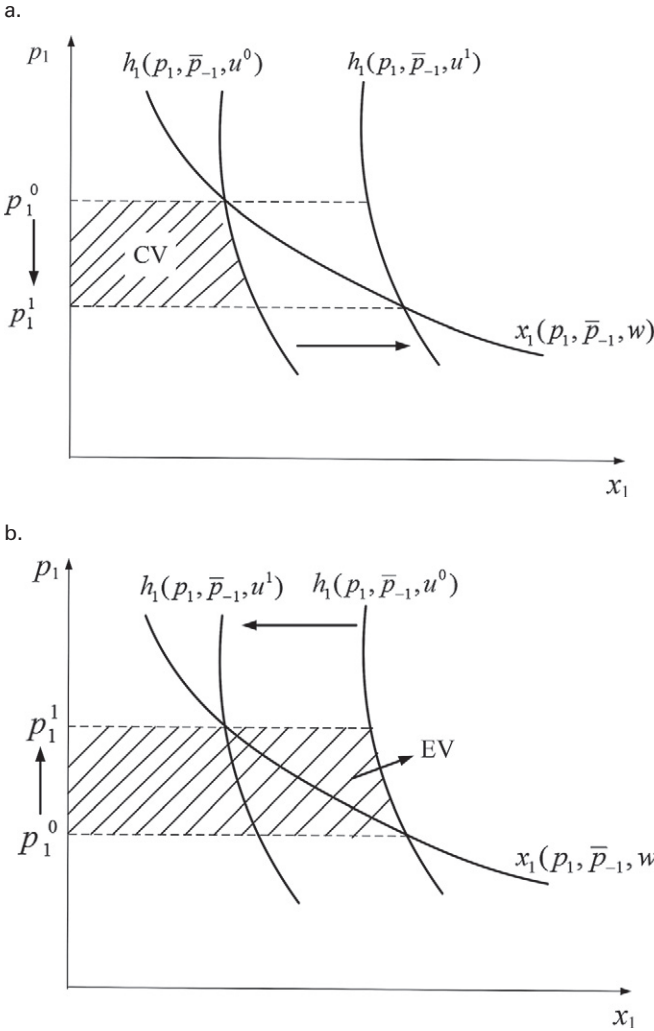


Figure 3.9
Normal goods, $EV > CS > CV$

$(25, 50)$, $x(p_1, w) = (50, 50)$, and $x(p_1, w') = (35.35, 35.35)$. Hence, the CV is given by the difference in income that the individual needs to purchase these two bundles, as follows, $CV = 100 - 70.71 = 29.29$. Last, to measure the EV , we need to obtain bundle $x(p_0, w'')$ where w'' represents the adjusted wealth so the individual can reach his final utility level before the price change. It is easy to show that $x(p_0, w'') = (35.35, 70.71)$. As a consequence, the EV is given by the difference in incomes from purchasing these two bundles, as follows, $EV = 141.42 - 100 = 41.42$. In summary, $EV > CS > CV$. For a numerical example in which, in contrast, the change in welfare yields the same measure when using CV , EV , and CS , consider a quasi-linear utility function $u(x, y) = 2\sqrt{x} + y$, still assuming the same wealth and prices as above. You should find that the Walrasian demand for good x in this context is $x(p_x, 100) = (1/p_x)^2$, yielding an increase in CS of 0.5. It is straightforward to show that the CV and EV produce the same amount, entailing $EV = CS = CV = \$0.50$. ■

In figure 3.10 we show the difference between these welfare measures for the case in which x_1 is inferior, namely income effects are negative. Importantly, the previous ranking between welfare measures is reversed, since $CV > CS$ and $EV < CS$, thus implying that $EV < CS < CV$. We can therefore summarize our previous findings in table 3.1, which represents the consequences of using consumer surplus as a measure of welfare change (rather than using the CV or EV).

Example 3.2: Finding CV , EV , and CS associated to a Cobb–Douglas utility function Consider a consumer with Cobb–Douglas utility function $u(x_1, x_2) = x_1^\alpha x_2^\beta$. From the parametric examples developed throughout chapter 2, we know that the Walrasian demands associated with this utility function are

$$x_1(p, w) = \frac{\alpha w}{(\alpha + \beta)p_1} \quad \text{and} \quad x_2(p, w) = \frac{\beta w}{(\alpha + \beta)p_2}.$$

Assume that the consumer's wealth is $w = \$10$, and prices are $p_1 = p_2 = 2$. For simplicity, consider parameters $\alpha = \beta = 1/2$. In this context the indirect utility function (also found in chapter 2) is

Table 3.1
Comparison of CV , EV , and CS

	Normal goods	Inferior goods
Price decrease	$EV > CS > CV$	$EV < CS < CV$
Price increase	$EV < CS < CV$	$EV > CS > CV$

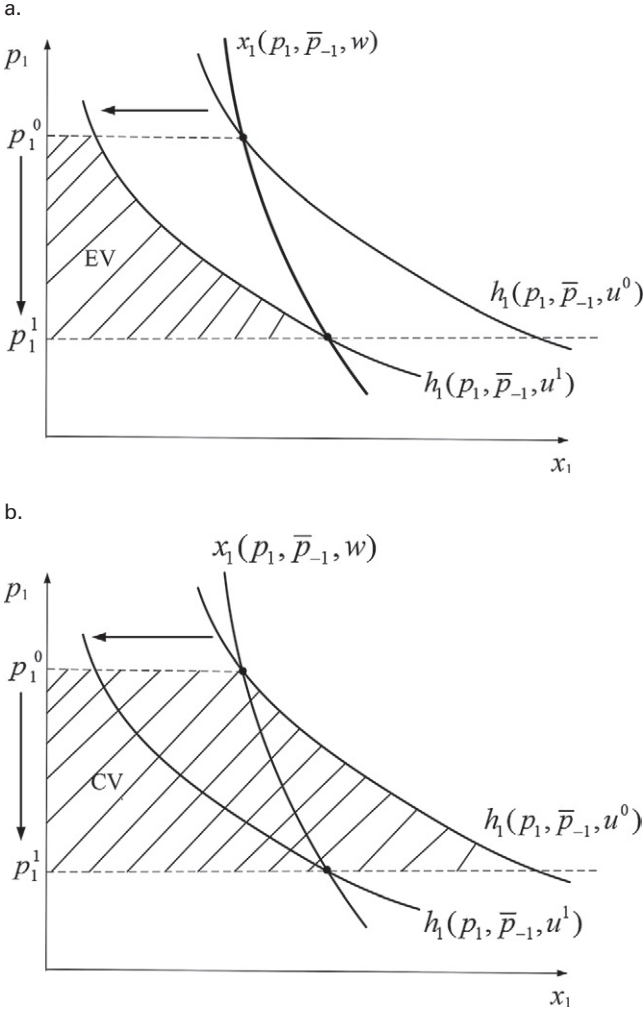


Figure 3.10
Inferior goods, $EV < CS < CV$

$$\begin{aligned}
v(p, w) &= \left(\frac{\alpha}{p_1}\right)^\alpha \left(\frac{\beta}{p_2}\right)^\beta \left(\frac{w}{\alpha + \beta}\right)^{\alpha + \beta} \\
&= \left(\frac{1}{2 \times 2}\right)^{1/2} \times \left(\frac{1}{2 \times 2}\right)^{1/2} \times \left(\frac{10}{1}\right)^1 \\
&= \frac{1}{2} \times \frac{1}{2} \times 10 = 2.5.
\end{aligned}$$

Hence, if the price of good 1 decreases by 50 percent, namely from $p_1 = 2$ to $p'_1 = 1$, the change in consumer surplus is

$$\begin{aligned}
CS &= \int_1^2 x_1(p, w) dp_1 \\
&= \int_1^2 \frac{\alpha w}{(\alpha + \beta) p_1} dp_1 \\
&= \int_1^2 \frac{5}{p_1} dp_1 = 5 \ln p_1 \Big|_1^2 = 5 \ln 2 \approx 3.466.
\end{aligned}$$

Let us now examine the compensating variation associated with this price decrease. First, recall that the Hicksian demands (found in chapter 2) are

$$h_1(p, u) = \left(\frac{\alpha p_2}{\beta p_1}\right)^{\beta/(\alpha + \beta)} u^{1/(\alpha + \beta)} \quad \text{and} \quad h_2(p, u) = \left(\frac{\beta p_1}{\alpha p_2}\right)^{\alpha/(\alpha + \beta)} u^{1/(\alpha + \beta)}.$$

Evaluating them at the utility level of the initial prices $p_1 = p_2 = 2$ found above, $u^0 = 2.5$, and parameters $\alpha = \beta = 1/2$, we obtain

$$\begin{aligned}
h_1(p, u^0) &= \left(\frac{\alpha p_2}{\beta p_1}\right)^{\beta/(\alpha + \beta)} (u^0)^{1/(\alpha + \beta)} = \left(\frac{\frac{1}{2} \times 2}{\frac{1}{2} \times 2}\right)^{1/2} \times 2.5 = 2.5, \\
h_2(p, u^0) &= \left(\frac{\beta p_1}{\alpha p_2}\right)^{\alpha/(\alpha + \beta)} (u^0)^{1/(\alpha + \beta)} = \left(\frac{\frac{1}{2} \times 2}{\frac{1}{2} \times 2}\right)^{1/2} \times 2.5 = 2.5.
\end{aligned}$$

Hence the compensating variation associated to a 50 percent decrease in the price of good 1 is

$$\begin{aligned}
CV &= \int_1^2 h_1(p, u^0) dp_1 \\
&= \int_1^2 \left(\frac{\alpha p_2}{\beta p_1} \right)^{\beta/(\alpha+\beta)} (u^0)^{1/(\alpha+\beta)} dp_1 \\
&= \int_1^2 \left(\frac{2}{p_1} \right)^{1/2} 2.5 dp_1 = 10 - 5\sqrt{2} \approx 2.929.
\end{aligned}$$

In words, the consumer would need to give up \$2.929 to be as well off as before the price decrease. Let us finally identify the equivalent variation of this price change. In order to do this, we first need to evaluate the indirect utility function $v(p, w)$ at the final prices $p_1 = 1$ and $p_2 = 2$, which yields $u^1 = 5/\sqrt{2} = 3.536$. Plugging this utility level u^1 in the Hicksian demand for good 1 (also evaluated at final prices $p_1 = 1$ and $p_2 = 2$), we obtain

$$h_1(p, u^1) = \left(\frac{\alpha p_2}{\beta p_1} \right)^{\beta/(\alpha+\beta)} (u^1)^{1/(\alpha+\beta)} = \left(\frac{\frac{1}{2} \times 2}{\frac{1}{2} \times 1} \right)^{1/2} \times \frac{5}{\sqrt{2}} = 5.$$

Hence the equivalent variation of the price decrease is

$$\begin{aligned}
EV &= \int_1^2 h_1(p, u^1) dp_1 \\
&= \int_1^2 \left(\frac{\alpha p_2}{\beta p_1} \right)^{\beta/(\alpha+\beta)} (u^1)^{1/(\alpha+\beta)} dp_1 \\
&= \int_1^2 \left(\frac{2}{p_1} \right)^{1/2} \times 3.536 dp_1 = 20 - 10\sqrt{2} \approx 5.858.
\end{aligned}$$

Intuitively, the consumer would need \$5.858 to be as well off as after the price decrease. ■

3.4 When Can We Use the Walrasian Demand as a Measure of Welfare Change?

Our previous discussion suggests that CS yields either a larger or smaller measure of welfare change than the CV and EV . Under certain cases, however, such measurement difference is relatively small, and therefore the use of the CS may be justified. This is especially valid when income effects associated to a price change are small.¹⁰ The use

¹⁰ This implies that graphically the Walrasian and the Hicksian demand curves almost coincide. The welfare change using the CV and EV also coincides. This coincidence arises, for instance, when utility is quasi-linear and thus income effects are absent.

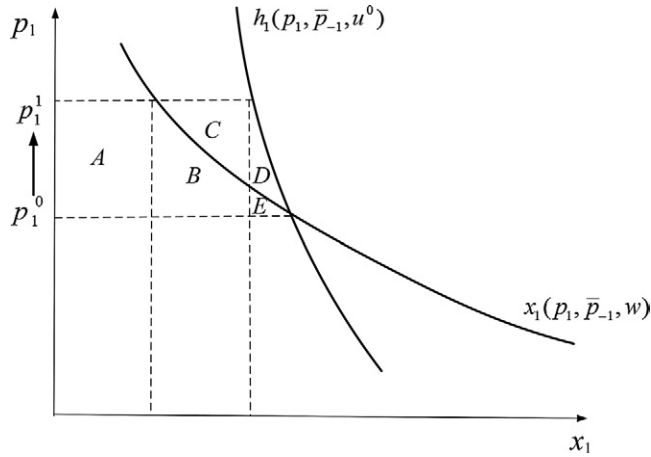


Figure 3.11
Small price change (using AV)

of the Walrasian demand is justified if, in addition, the price change is small. Indeed the measurement difference in using the CS, as a fraction of the welfare change (measured by the CV), becomes small. Figure 3.11 illustrates this point, where the welfare change according to CS is $A + B + E$, while that from using the CV is $A + B + C + D + E$. Hence the measurement difference is given by $C + D$. Specifically, as the price increase becomes negligible, area $C + D$ as a fraction of the compensating variation gets close to zero as well. That is,

$$\lim_{(p_1^1 - p_1^0) \rightarrow 0} \frac{C + D}{CV} = \lim_{(p_1^1 - p_1^0) \rightarrow 0} \frac{C + D}{A + B + C + D + E} = 0,$$

since the compensating variation is $CV = A + B + C + D + E$.

Another option when the price change is relatively small is to take a first-order Taylor approximation¹¹ of the Hicksian demand, $h_1(p, u^0)$, which we denote as $\tilde{h}_1(p, u^0)$, at the initial price vector, p^0 . That is,

$$\tilde{h}(p, u^0) = h(p^0, u^0) + D_p h(p^0, u^0)(p - p^0).$$

We then calculate the CV of this “approximated” Hicksian demand, $\tilde{h}_1(p, u^0)$, the same as we did the “true” Hicksian demand:

¹¹ For an introduction to first-order Taylor approximations, see section 30.2 in Simon and Blume’s (1994) textbook.

$$CV = \int_{p_1^0}^{p_1^1} \tilde{h}_1(p_1, \bar{p}_{-1}, u^0) dp_1,$$

where, since $h(p^0, u^0) = x(p^0, w)$ and $D_p(p^0, u^0) = S(p^0, w)$, we can rewrite the “approximated” Hicksian as

$$\tilde{h}(p, u^0) = x(p^0, w) + S(p^0, w)(p - p^0).$$

Figure 3.12 illustrates our results. First, the “approximated” Hicksian demand $\tilde{h}_1(p^0, u^0)$ lies between the true Hicksian demand, $h_1(p_1, \bar{p}_{-1}, u^0)$, and the Walrasian demand, $x(p^0, w)$. In particular, the expression above indicates that $\tilde{h}_1(p, u^0)$ takes the same value as the Walrasian demand at the initial price level p^0 and that it has the same slope as the true Hicksian demand, $h_1(p_1, \bar{p}_{-1}, u^0)$, at the initial price level p^0 measured by $S(p^0, w)$, since the slope of the Hicksian demand, $D_p h(p^0, u^0)$, satisfies $D_p h(p^0, u^0) = S(p^0, w)$. The “approximated” Hicksian demand $\tilde{h}_1(p, u^0)$ has, however, a different slope than the true Hicksian as we move away from price p_1^0 . Nonetheless, if the price change is sufficiently small, the compensating variation measured using the “approximated” Hicksian demand $\tilde{h}_1(p, u^0)$ will not imply a substantial error. The compensating variation is indicated in figure 3.12, as well as the measurement error.

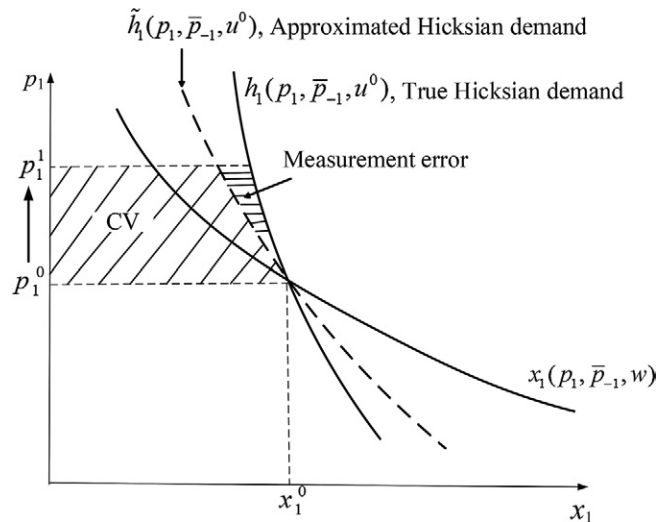


Figure 3.12
“Approximated” Hicksian demand, $\tilde{h}_1(p_1, \bar{p}_{-1}, u^0)$

3.5 Application of Income and Substitution Effects—I

In this section we describe income and substitution effects in a compact and tractable expression that allows for empirical applications. Let us first represent the Slutsky equation using a more compact notation. First, from the Slutsky equation we know that

$$\frac{\partial h_1(p, u)}{\partial p_1} = \frac{\partial x_1(p, w)}{\partial p_1} + \frac{\partial x_1(p, w)}{\partial w} x_1(p, w).$$

Multiplying both sides by p_1/x_1 obtains

$$\frac{\partial h_1(p, u)}{\partial p_1} \frac{p_1}{x_1} = \frac{\partial x_1(p, w)}{\partial p_1} \frac{p_1}{x_1} + \frac{\partial x_1(p, w)}{\partial w} x_1(p, w) \frac{p_1}{x_1}.$$

And multiplying the last term by $w/w=1$, yields

$$\underbrace{\frac{\partial h_1(p, u)}{\partial p_1} \frac{p_1}{x_1}}_{\text{Substitution price elasticity of demand, } \tilde{\epsilon}_{p,Q}} = \underbrace{\frac{\partial x_1(p, w)}{\partial p_1} \frac{p_1}{x_1}}_{\text{Price elasticity of demand, } \epsilon_{p,Q}} + \frac{\partial x_1(p, w)}{\partial w} x_1(p, w) \frac{p_1}{x_1} \frac{w}{w}.$$

We can rearrange the last term of the expression above, as a product of the income elasticity of demand and the share of budget that the consumer spends on good 1, as follows:

$$\frac{\partial x_1(p, w)}{\partial w} x_1(p, w) \frac{p_1}{x_1} \frac{w}{w} = \underbrace{\frac{\partial x_1(p, w)}{\partial w} \frac{w}{x_1}}_{\text{Income elasticity of demand, } \epsilon_{w,Q}} \underbrace{\frac{p_1 x_1(p, w)}{w}}_{\text{Share of budget spent on good 1, } \theta}.$$

We can therefore more completely rewrite the Slutsky equation in terms of elasticities as

$$\tilde{\epsilon}_{p,Q} = \epsilon_{p,Q} + \theta \epsilon_{w,Q}.$$

Let us consider an example in order to provide intuition behind this equation. Consider a good like housing, with a budget share of 40 percent for many families.¹² Assume that the income elasticity of the demand for housing is 1.38 and the price elasticity of the demand for housing is -0.6 . Thus

¹² A typical household's housing budget is 33 to 38 percent (59 to 66 percent of this is on shelter—mortgage interest, property taxes, repairs, and rent, and other items); see <http://www.eioba.com/a9036/>.

$$\tilde{\epsilon}_{p,Q} = \epsilon_{p,Q} + \theta\epsilon_{w,Q} = -0.6 + (0.4 \times 1.38) = -0.05.$$

Intuitively this result implies that, if the price of housing increases by 10 percent, and consumers do not receive a wealth compensation to maintain their welfare unchanged, consumers will reduce their consumption of housing by 6 percent, that is, $\epsilon_{p,Q} = -0.6$. However, if consumers receive such (Hicksian) wealth compensation, the reduction in their housing consumption will only be of 0.5 percent, that is, $\tilde{\epsilon}_{p,Q} = -0.05$.¹³

Another immediate lesson from the Slutsky equation above is that price elasticities (with and without compensation) are close if the budget share spent on a particular good, θ , is very small (e.g., garlic), and/or if the income elasticity of this good is really small (i.e., pizza or Spam).¹⁴ In this case the Walrasian and Hicksian demand are very close to each other, implying that the welfare changes measured by CV , EV , and CS are all relatively similar. A similar argument applies when consumer's preferences are represented with a quasi-linear utility since income effects are absent.

Example 3.3: Empirical studies using CV to measure welfare changes In 1981 the US government negotiated a voluntary automobile export restriction program with the Japanese government. In 1987, Clifford Winston studied the effects of this program. First, he found that the average price of a Japanese (US) car went up by 20 percent (8 percent, respectively) as a result of the restrictions. What was the effect of these higher prices on consumers' welfare? Using the CV he found that $CV = -\$14$ billion. Using the definition of the CV , we can provide an intuitive interpretation as this result: the wealth compensation that domestic car owners would need after the price change (after the export restriction program) in order to be as well off as they were before the price change is \$14 billion. In another study, Jerry Hausmann also uses the CV to measure welfare changes. In this case, he measured the welfare gain consumers experience from the price reduction associated to the opening of a new Walmart store in their locality/county. The author used CV because low-income families spend a non-negligible share of their budget in Walmart stores, meaning a high value of θ . He found

¹³ In other words, housing is such an important share of most families' expenses that higher prices lead them to significantly reduce their consumption (if they are not compensated after the price increase), but just slightly do so if they were compensated.

¹⁴ If elasticities satisfy $\tilde{\epsilon}_{p,Q} \approx \epsilon_{p,Q}$, then $\theta\epsilon_{w,Q}$ must be close to zero (i.e., either θ or $\epsilon_{w,Q}$, or both, are small.).

that the opening of a new Walmart store represents an average increase of 3.75 percent in consumer welfare.¹⁵ ■

3.6 Application of Income and Substitution Effects—II: The Consumer as a Labor Supplier

In this section we apply our analysis of the income and substitution effects to an individual's decision about how many hours to work. In particular, this individual enjoys consumption of all other goods, x (a vector of N different goods), and leisure hours, L , and his UMP can be expressed as follows:

$$\max_{x,L} u(x, L) \quad \text{subject to} \quad \sum_{i=1}^K p_i x_i \leq M,$$

where $M = Z + \bar{M}$. M is the individual's total wealth coming from two sources: the z hours he dedicates to work (paid at a wage rate of w per hour), $Z = zw$, and from his nonlabor income $\bar{M}$, such as inheritance and government subsidies. (Note that his total time, $T = z + L$, must be dedicated to either work, z , or leisure, L).

We can rewrite the UMP above using the composite commodity theorem, as follows: if the prices of all goods maintain a constant proportion with respect to the price of labor (wages), $p_1 = \alpha_1 w, p_2 = \alpha_2 w, \dots, p_n = \alpha_n w$, then we can represent these goods by a single (composite commodity) y with price p . This is useful because, when we examine many goods, the relationships between all the demands can become very complicated. It is useful to be able to group the goods together and then examine only the remaining good. We therefore collapse the UMP above to only two goods: the composite commodity y and the number of hours dedicated to work z . That is, we can rewrite the UMP as

$$\max_{y,z} v(y, z) \quad \text{subject to} \quad py \leq wz + \bar{M},$$

where py denotes the amount of money spent on consumption goods (embodied in the composite commodity y) and $wz + \bar{M}$ describes the total income originating from labor sources, wz , and nonlabor sources, $\bar{M}$. The Lagrangian associated to this UMP is

$$L = v(y, z) + \lambda(\bar{M} + wz - py),$$

¹⁵ You can read more about this study in: <http://econ-www.mit.edu/files/1036>. For more detail on the use of the compensating and equivalent variation as a measure of welfare change, see McKenzie (1983) and, chapters 7–10 in Boadway and Niel (1984); see also Johansson (1991).

and the first-order conditions (for an interior optimum) are

$$\begin{aligned}\frac{\partial L}{\partial y} : v_y - \lambda p = 0 &\Rightarrow \lambda = \frac{v_y}{p}, \\ \frac{\partial L}{\partial z} : v_z + \lambda w = 0 &\Rightarrow \lambda = -\frac{v_z}{w}.\end{aligned}$$

Since both expressions are equal to λ , we can set them equal to each other to obtain $v_y/p = -v_z/w$. Intuitively, the individual allocates hours in a day so that the marginal utility per dollar spent on consumption goods is equal to the marginal utility of working hours per dollar earned. Using the constraint, we finally obtain the Walrasian demand for the composite commodity $x_y(w, p, \bar{M})$ and the labor supply function $x_z(w, p, \bar{M})$. Figure 3.13 illustrates the individual's optimal choice of the composite commodity, y , and hours of work, z .

First, note that the budget line is represented by an upward-sloping straight line. Intuitively, an increase in the amount of hours worked provides the individual with a larger amount of wealth to spend on consumption goods, measured by the composite commodity in the vertical axis.¹⁶ Second, note that this individual's indifference curves show increasing utility levels as we move northwest, indicating that the individual is better off when his consumption of the composite commodity increases and the number of working hours decreases. For a starting wage rate w , this individual selects bundle A , where he works z_0 hours. When the wage rate increases to w_1 the budget line becomes steeper (i.e., for every extra hour of work the individual can afford a larger amount of the composite commodity). At this new wage rate w_1 the individual chooses bundle B , spending z_1 hours working. The change from bundle A to B in the top panel of figure 3.13 is also reflected in the bottom panel of figure 3.13, which depicts the labor supply curve. For an increase in the wage rate from w_0 to w_1 , the number of working hours increases from z_0 to z_1 , indicating that a higher wage induces this worker to spend more hours on the job. When the wage rate experiences a further increase, from w_1 to w_2 , the budget line becomes steeper in the top figure, and the individual selects bundle C , illustrating a *reduction* in the number of working hours, from z_1 to z_2 . This effect is also reflected in the bottom figure, where an increase in the wage rate from w_1 to w_2 induces

¹⁶ In particular, solving for y in $py = wz + \bar{M}$, the equation of the budget line becomes $y = z(w/p) + \bar{M}/p$. Therefore the budget line originates at $\bar{M}/p$, which indicates the amount of the composite commodity the consumer can afford when he does not work, while the positive slope of the individual's budget line is w/p . Additionally note that indifference curves reflect the fact that preferences are quasi-concave.

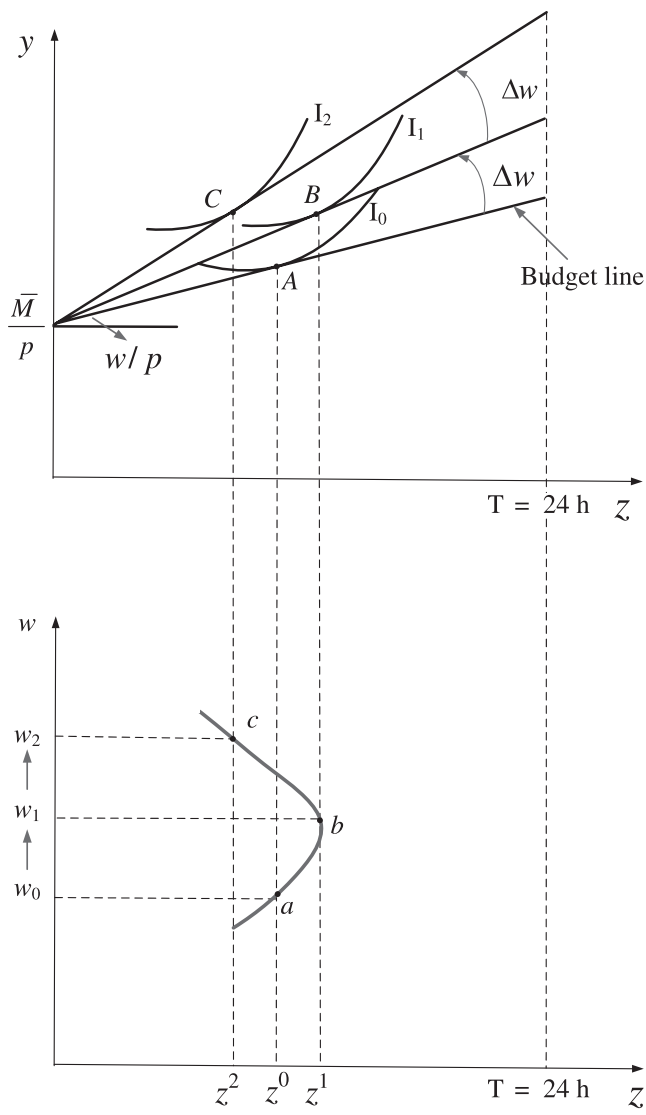


Figure 3.13
Consumer as a labor supplier

a reduction in the individual's labor supply curve. Summarizing, the labor supply initially increases as a result of higher wages but then decreases, exhibiting as a result the backward bending pattern of the bottom figure.¹⁷ This is due to the relative size of the substitution and income effect associated to the increase in the wage rate (a change in the price of one of the goods that the individual consumes), as we examine next. Figure 3.14 illustrates the substitution and income effects from a wage increase.

Using the same analysis as for two consumption goods, we make a wealth compensation *after* the price change that leaves the consumer as well off as *before* the price change. This is indicated in figure 3.14 by the downward shift in the budget line after the price change toward a new budget line (dashed line in the figure), which is tangent to the indifference curve the individual reaches before the price change, I_1 . Given this new budget line, the individual selects bundle D . The substitution effect of an increase in the wage rate is hence measured by an increase in working hours, from z_A to z_D , while the income effect is represented by a decrease in the number of working hours, from z_D to z_B . Intuitively, working hours become more attractive (relative to

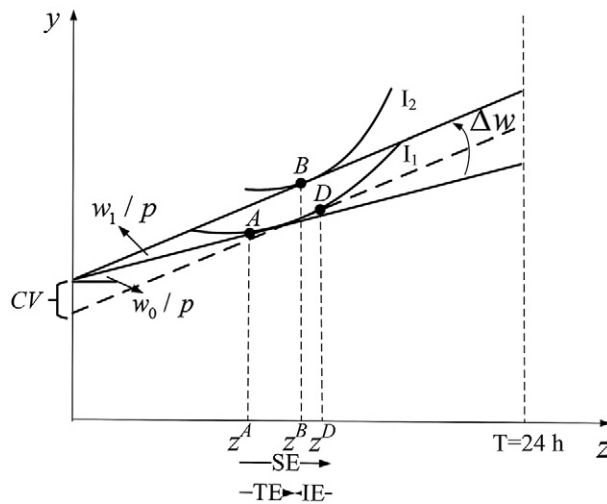


Figure 3.14
Income and substitution effects in the labor market

¹⁷ This effect has been empirically confirmed in some occupations, such as nursing services in Massachusetts. Experiencing a shortage in the number of nurses, managers of hospital facilities decided to increase the wage per hour in order to attract more nurses. Unfortunately, the increase in wages was counterproductive, at least in the short run, since it induced nurses currently working for those hospitals to reduce the number of hours they chose to work. In the long run, however, a few more nurses were attracted to the area by the high wages. (For more details, see Spetz et al. 2005.)

the composite good) for the individual, leading him to offer more labor services, as reflected in the substitution effect. This higher wage per hour, however, allows the individual to afford more consumption without the need to work so many hours per day, which induces him to reduce his working hours, as indicated by the negative income effect. For the indifference curves depicted in figure 3.14, the income effect partially offsets the substitution effect, leading to a relatively minor (but still positive) total effect. When the income effect is negative and, in absolute value, larger than the substitution effect, the total effect of a higher wage rate becomes negative.¹⁸

Figure 3.14 also illustrates the compensating variation (CV) associated with the wage increase. In particular, after the wage increase the worker's wealth is compensated (reduced) so that he can maintain his initial utility level (before the wage increase).¹⁹ Graphically, we do so by shifting the worker's budget line downward after the wage increase in a parallel fashion (maintaining the final price ratio) until the worker reaches his initial utility level at the indifference curve I_1 . This wealth compensation represents the CV . In addition, recall that the vertical intercept of the worker's initial budget line is $\bar{M}/p$, while the vertical intercept of his final budget line is M'/p , where M' represents the wealth level that the individual needs in order to reach his initial utility level at the final price ratio, $M'/p < \bar{M}/p$. Therefore the amount of money that the individual is willing to give up in order to reach his initial utility level at the final price ratio (i.e., when wages are higher) is measured by the difference $(\bar{M}/p) - (M'/p)$. This difference represents the CV of the wage increase, and it is graphically depicted in the vertical axis of figure 3.14. Hence, if the price of all other goods is normalized to one, meaning $p = \$1$, the difference in the vertical intercepts $\bar{M} - M'$ represents the compensating variation.

3.6.1 The Slutsky Equation in Labor Markets

Having described the income and substitution effect in the labor market from a graphical approach, let us formalize these two effects using the Slutsky equation. First, consider the EMP associated with the previous problem:

$$\min_{y,z} M = py - wz \quad \text{subject to} \quad v(y, z) = v,$$

¹⁸ In such a case, working hours decrease as a result of an increase in the wage rate, becoming a Giffen good. This is the case represented in the backward-bending portion of the labor supply curve described above, for which an increase in the wage rate induces a net reduction in the working hours.

¹⁹ Recall that the CV is the amount of money that must be taken from the consumer in the new situation (after the increase in the wage rate w) to make him as well off as he was in the initial situation (before the increase in wages).

where the worker seeks to minimize his expenditure while targeting a given utility level v . From this EMP, we can find the optimal Hicksian demands, $h_y(w, p, v)$ and $h_z(w, p, v)$, for consumption of the composite commodity, y , and working hours, z , respectively. Inserting them into the objective function, we obtain the value function of this EMP (the expenditure function):

$$e(w, p, v) = ph_y(w, p, v) + wh_z(w, p, v),$$

where $x_z(w, p, e(w, p, v)) = h_z(w, p, v)$. Differentiating both sides and using the chain rule obtains

$$\frac{\partial x_z}{\partial w} + \frac{\partial x_z}{\partial e} \frac{\partial e}{\partial w} = \frac{\partial h_z}{\partial w} \Leftrightarrow \frac{\partial x_z}{\partial w} = \frac{\partial h_z}{\partial w} - \frac{\partial x_z}{\partial e} \frac{\partial e}{\partial w}.$$

And since the last term on the right-hand side can be rewritten as $-\partial e(w, p, v) / \partial w = -h_z(w, p, v)$, then

$$\underbrace{\frac{\partial x_z}{\partial w}}_{SE} = \underbrace{\frac{\partial h_z}{\partial w}}_{TE} + \underbrace{\frac{\partial x_z}{\partial e}}_{IE} h_z(w, p, v).$$

Let us next interpret the Slutsky equation above in terms of the substitution and income effect in the labor market. First, term $\partial h_z / \partial w$ represents the substitution effect. It is always positive, indicating that an increase in wages increases the worker's supply of labor (as long as we compensate the wealth of the worker so that his initial utility level is unaffected²⁰). The second term, $(\partial x_z / \partial e) h_z(w, p, v)$, denotes the income effect. When $\partial x_z / \partial e > 0$, an increase in wages makes the worker richer, and he decides to work more. In this case working hours are regarded as a normal good, and the income effect reinforces the substitution effect yielding an upward-sloped labor supply curve. If, in contrast, $\partial x_z / \partial e < 0$, an increase in wages makes the worker richer, but he decides to work less. In this case, working hours are regarded as an inferior good, and the (negative) income effect moves in the opposite direction of the substitution effect. When the income effect is relatively small, the total effect of an increase in wages is still positive, producing a positively sloped labor supply curve. If, however, the income effect is sufficiently negative, the (negative) income effect completely offsets the (positive) substitution effect, implying that the total effect of an increase in wages becomes negative, yielding a negatively sloped labor supply curve.

²⁰ This implies reducing the worker's wealth in the case of a wage increase (as in our previous example), or increasing it in the case of a wage decrease.

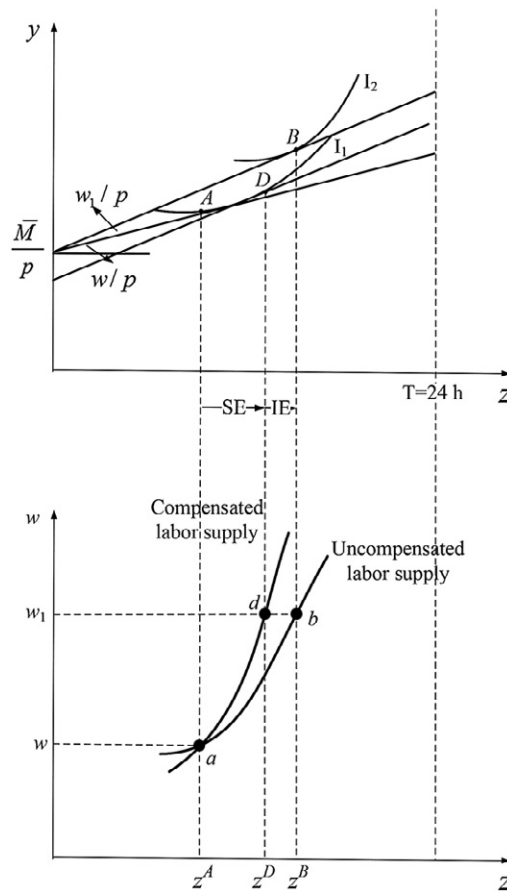


Figure 3.15
Compensated and uncompensated labor supply curves

Figure 3.15 illustrates a case in which the income effect resulting from a wage increase is positive, yielding a positively sloped labor supply curve for all wages. For completeness, we also include the compensated labor supply curve, which captures the substitution effect due to the wage increase but not the income effect. As a consequence the compensated supply curve is always positively sloped because the substitution effect of a wage increase is positive. The uncompensated labor supply curve, in contrast, represents both the substitution and income effect. As a consequence it is positively sloped when the income effect is positive, or when, despite being negative, the income effect is smaller (in absolute value) than the substitution effect. Graphically, that would entail a positively sloped uncompensated labor supply curve but,

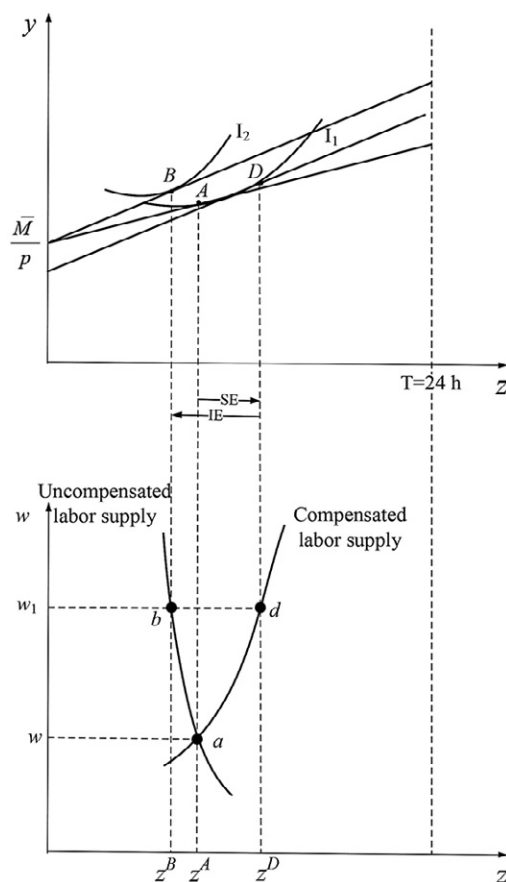


Figure 3.16
Negatively sloped uncompensated labor supply curve

unlike in figure 3.15, lying below the compensated labor supply curve. When the income effect is negative, and more than offsets the substitution effect, the uncompensated supply curve becomes negatively sloped as figure 3.16 illustrates.²¹

²¹ From an empirical point of view, there is substantial evidence showing that the labor supply curve for British, American, Japanese, and Dutch men is virtually vertical, indicating that the substitution and income effects essentially offset each other; the labor supply curve for British and German married women is almost vertical, suggesting a similar ranking between substitution and income effects; the labor supply curve for US and Canadian married women is slightly backward bending, indicating that the income effect becomes significantly negative for relatively high wages; and the supply curve for single women in most developed countries is positively sloped, indicating that the income effect is not significantly negative (i.e., the elasticity of labor supply of single women is larger than 4 in many studies).

3.6.2 The Laffer Curve

An interesting application of the income and substitution effects in the labor market is related with the so-called Laffer curve. The supporters of this curve argue that increasing the tax rate on wages might initially increase tax revenue, but after a certain tax rate, further increments only reduce the incentives to work, ultimately reducing tax collection as well. This suggests that there is an optimal tax rate τ^* that will bring in the most tax revenue, while any increase or decrease from that point will cause a reduction in tax collection. Graphically, the Laffer curve resembles an inverted-U, with tax rates on the horizontal axis and tax revenue on the vertical axis, as depicted in figure 3.17.

In order to understand the precise relationship between tax rate and tax revenue in this setting, let us define w as the nominal wage (before taxes), $\omega = (1 - \tau)w$ as the real wage (after taxes), and let $H(\omega)$ denote the total number of working hours supplied when the real wage is ω . Hence tax revenue is $T = \tau \cdot w \cdot H(\omega)$. Differentiating with respect to the tax rate, we obtain

$$\frac{\partial T}{\partial \tau} = \underbrace{w \cdot H(\omega)}_{\text{Positive effect}} - \underbrace{\tau w^2 \frac{dH}{d\omega}}_{\text{Negative effect}}.$$

Intuitively, the first term represents a positive (direct) effect of higher tax rates, since for a given number of working hours, $H(w)$, a higher tax rate increases total revenue. The second term denotes, however, a negative (indirect) effect from increasing the tax

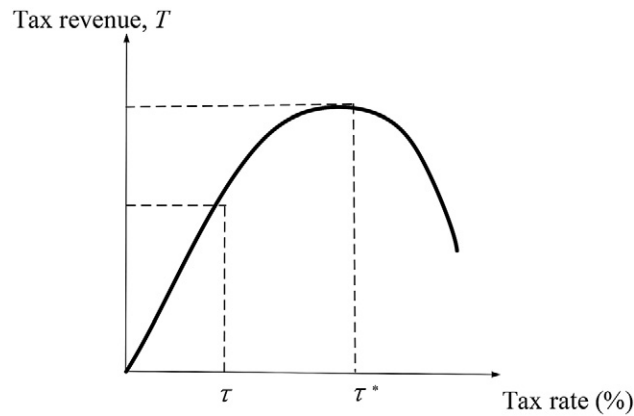


Figure 3.17
Laffer curve

rate τ . In particular, higher taxes induce workers to work fewer hours, decreasing the tax revenue.²² Therefore the negative effect dominates the positive effect (and as a consequence an increase in the tax rate produces an overall reduction in tax revenue) if the expression above is negative, that is,

$$w \cdot H(w) < \tau w^2 \frac{dH}{dw} \quad \text{or} \quad \frac{1}{\tau} < \frac{dH}{d\omega} \cdot \frac{w}{H(\omega)}.$$

And multiplying both sides by $(1 - \tau)$ obtains

$$\frac{1 - \tau}{\tau} < \frac{dH}{d\omega} \frac{w(1 - \tau)}{H(\omega)},$$

where $\omega = w(1 - \tau)$ is the wage net of taxes, and therefore the term at the right-hand side of the inequality can be more compactly expressed as $(dH/d\omega) \cdot (\omega/H(\omega))$, which denotes the elasticity of labor supply with respect to net wages, $\varepsilon_{\text{supply}, \omega}$. Hence the condition above can be further simplified to

$$\frac{1 - \tau}{\tau} < \varepsilon_{\text{supply}, \omega}.$$

Therefore, for tax revenue to fall from a small increase in the tax rate, the elasticity of labor supply must be larger than $(1 - \tau)/\tau$, suggesting that τ would have to be sufficiently large. This condition, however, is relatively difficult to be satisfied. For instance, if the marginal tax rate²³ for the most affluent citizen is $\tau = 0.80$, then the condition above implies that $\varepsilon_{\text{supply}, \omega} > (1 - 0.8)/0.8 = 0.25$, which is very likely to be satisfied. Nonetheless, an economy in which the maximum marginal tax rate is $\tau = 0.35$ would need the elasticity of labor supply should be larger than $(1 - \tau)/\tau = (1 - 0.35)/0.35 = 1.86$, for total tax revenue to increase, which is very unlikely to hold for the average worker in most developed countries. Figure 3.18 depicts cutoff $(1 - \tau)/\tau$ for a given tax rate τ , the elasticity of labor supply, $\varepsilon_{\text{supply}, w}$, should be larger than the cutoff for condition $\varepsilon_{\text{supply}, w} > (1 - \tau)/\tau$ to hold (as depicted in the shaded area).

Further reading In a recent study, Trabandt and Uhlig (2011) compare tax rates in several developed economies with the tax rate that maximizes tax revenue according to the Laffer curve. Their results suggest that current tax rates are generally below

²² This implies that the uncompensated labor supply has a positive slope, whereby wage and working hours move in the same direction.

²³ Recall that the marginal tax rate represents the tax rate applicable to the additional dollar an individual earns, not to all of his/her income.

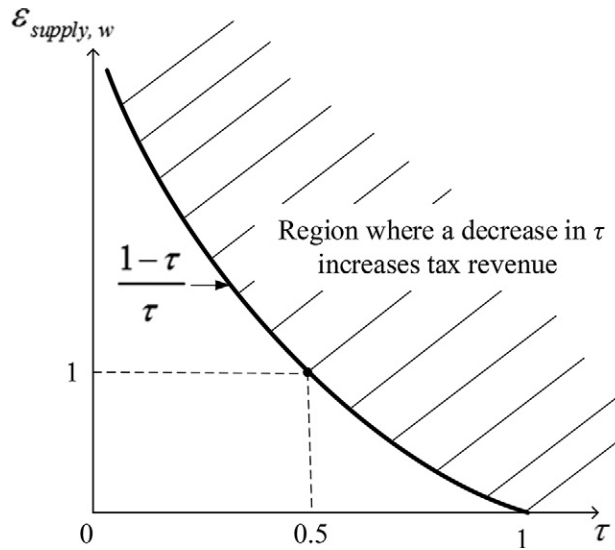


Figure 3.18
 $(\varepsilon_{\text{supply}, w}, \tau)$ -pairs for which the Laffer curve holds

those maximizing total tax collection, except in a few countries, like Denmark. (For more details, see Trabandt and Uhlig 2011.)

3.7 Application of Income and Substitution Effects—III: Income and Substitution Effects among Different Goods

In previous sections we focused on the substitution and income effects of varying the price of good k on the demand for that same good. In this section, however, we analyze the substitution and income effects of varying the price of good k on the demand for other good j . In terms of the Slutsky matrix, we paid close attention to the elements along the main diagonal of the matrix, whereas we now turn to the elements away from the main diagonal. For simplicity, let us start our analysis using only two goods, x and y . The types of relationships that can occur between only two goods are, of course, relatively limited, but this will help us graphically illustrate their main intuitions. Below we generalize our analysis to $N > 2$ goods. Let us start defining goods that are gross complements or substitutes in consumption. In particular, when p_y falls, the substitution effect can be so small that the consumer purchases a larger amount of both goods x and y . In this case we denote good x and y as *gross complements*. Figure 3.19 illustrates this case. Specifically, the consumer starts purchasing bundle A . Afterward,